MSHA Unseen Narrative Analysis

Block 2 - Causes, Contributing Factors, and Prevention Themes

Generated October 16, 2026 at 03:03

```
Unseen narratives processed: 2,000
Block 1 development IDs excluded: 1,000
Cause-analysis population: 1,654
Full-analysis population: 111
Excluded/review population: 346
Unsupported factors replaced: 761
Model: Qwen/Qwen2.5-7B-Instruct
Prompt: MSHA_BLOCK1C_FROZEN_FOR_BLOCK2_V3.0

Objective:
Evaluate how causes, contributing factors, and preliminary prevention
themes vary by injury severity and time period.

Temporal results represent changes in narrative-category shares.
They are not exposure-adjusted injury-incidence rates.
```

Executive Summary

Block 2 applies the frozen Block 1c model and ontology to 2,000 narratives that were not used during ontology development.

The sample contains 400 records from each of five injury-severity groups and is distributed across early, middle, and recent periods.

The model assigns a record type, primary cause, contributing factor, prevention theme, evidence excerpts, confidence values, and support levels.

A deterministic safeguard removes contributing factors that lack supporting narrative evidence. The factor is replaced with NO_SEPARATE_FACTOR_STATED and the prevention theme is changed to INVESTIGATION_REQUIRED.

Cause analysis includes valid causes even when no separate factor is supported. Factor and prevention analyses use only records valid for full analysis.

Within-group percentages are used for severity and time comparisons. Chi-square tests and Cramer's V summarize differences in category distributions but do not establish causation.

Sample by Severity and Time Period

Severity Group	Early: 2000-2008	Middle: 2009-2017	Recent: 2018-2026
Fatality	134	133	133
Permanent disability	134	133	133
Days away	134	133	133
Restricted or transferred	134	133	133
No lost time or other	134	133	133

Processing and Quality Summary

Measure	Value
Block 2 narratives	2000.0000
Successful classifications	1977.0000
Schema-valid classifications	1977.0000
Valid cause evidence	1786.0000
Unsupported-factor safeguards	761.0000
Factor-prevention aligned	1865.0000
Critical review required	377.0000
Advisory review	1790.0000
Valid for full analysis	111.0000
Valid for cause analysis only	1543.0000
Excluded or review required	346.0000
Mean processing seconds	18.0000

Record-Type Distribution

Record Type	Count	Percent
ACUTE_ACCIDENT	1940	97.0000
OCCUPATIONAL_EXPOSURE_OR_ILLNESS	29	1.4500
INSUFFICIENT_EVENT_INFORMATION	22	1.1000
NONWORK_RELATED_OR_NATURAL_EVENT	9	0.4500

Analysis-Status Distribution

Analysis Status	Count	Percent
VALID_FOR_CAUSE_ANALYSIS_ONLY	1543	77.1500
EXCLUDED_OR_REVIEW_REQUIRED	346	17.3000
VALID_FOR_FULL_ANALYSIS	111	5.5500

Critical Review Reasons

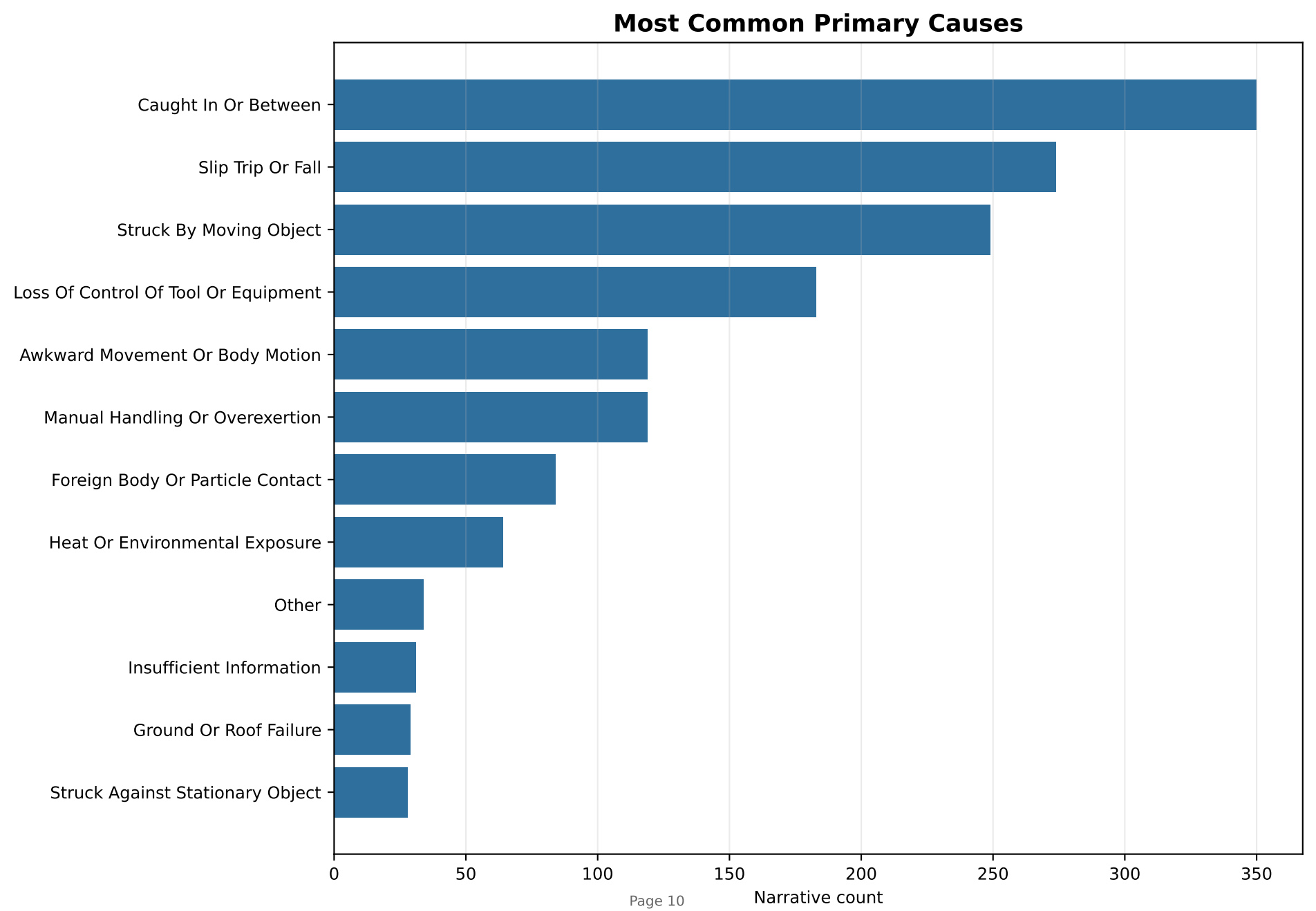
Critical Review Reason	Count
Cause evidence is not supported by narrative	214
Factor-prevention combination is outside allowed map	135
Cause is insufficient information	100
Record routed outside primary acute-accident population	60
Model response failed validation	23
One or more required fields were absent and defaulted	22

Advisory Review Reasons

Advisory Review Reason	Count
No separate contributing factor stated	1776
At least one support level is low	1764
At least one confidence value is low	1753
Unsupported factor conservatively replaced	761
Allowed alternative prevention used instead of preferred mapping	79
Narrative may be truncated	28

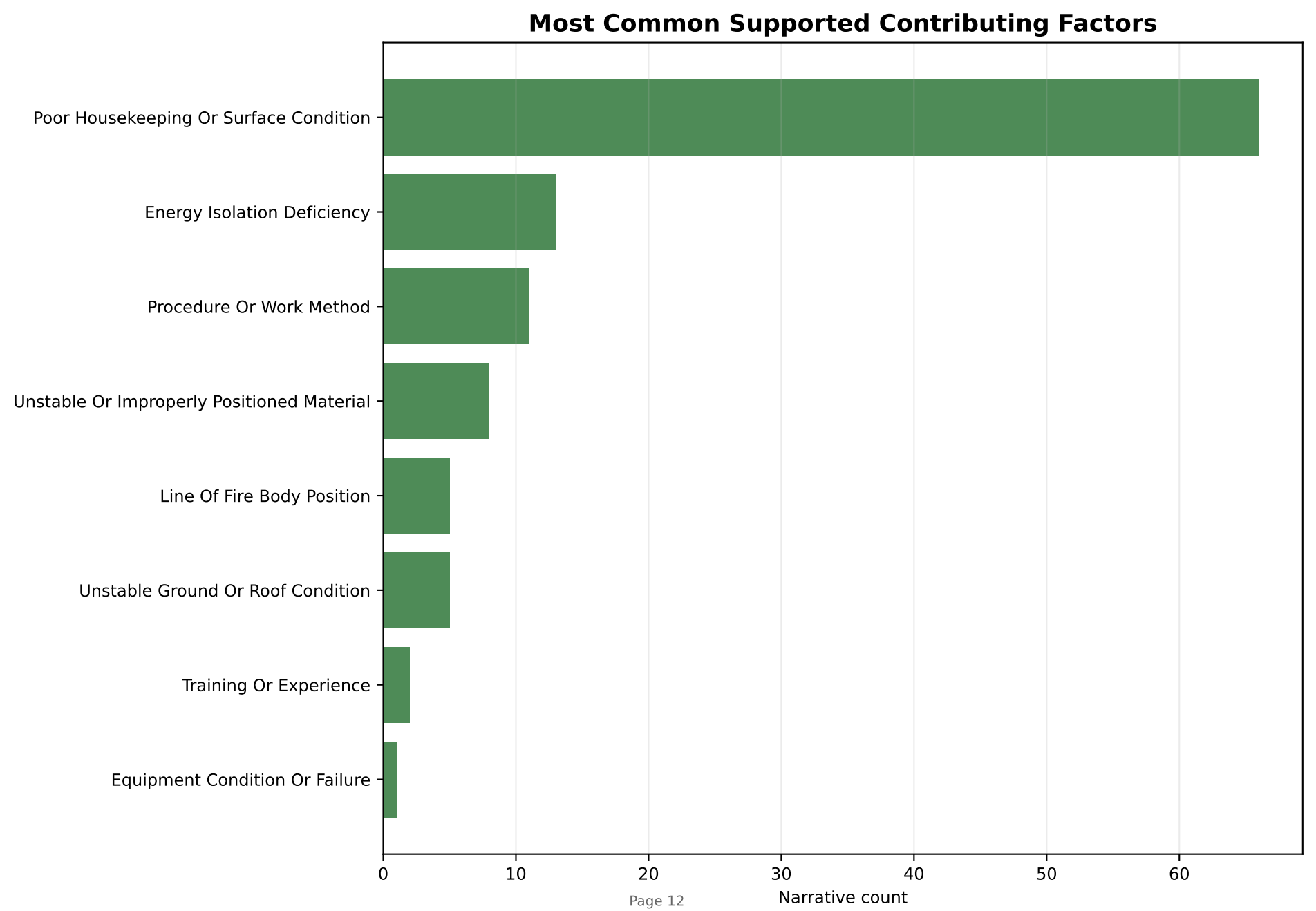
Primary Cause Distribution

Primary Cause	Count	Percent
CAUGHT_IN_OR_BETWEEN	350	21.1608
SLIP_TRIP_OR_FALL	274	16.5659
STRUCK_BY_MOVING_OBJECT	249	15.0544
LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT	183	11.0641
MANUAL_HANDLING_OR_OVEREXERTION	119	7.1947
AWKWARD_MOVEMENT_OR_BODY_MOTION	119	7.1947
FOREIGN_BODY_OR_PARTICLE_CONTACT	84	5.0786
HEAT_OR_ENVIRONMENTAL_EXPOSURE	64	3.8694
OTHER	34	2.0556
INSUFFICIENT_INFORMATION	31	1.8742
GROUND_OR_ROOF_FAILURE	29	1.7533
STRUCK_AGAINST_STATIONARY_OBJECT	28	1.6929
CHEMICAL_DUST_OR_OCCUPATIONAL_EXPOSURE	28	1.6929
PRESSURE_OR_STORED_ENERGY_RELEASE	21	1.2696
ELECTRICAL_CONTACT	18	1.0883
UNEXPECTED_EQUIPMENT_MOVEMENT	12	0.7255
VEHICLE_OR_MOBILE_EQUIPMENT_EVENT	6	0.3628
FIRE_EXPLOSION_OR_THERMAL_EVENT	4	0.2418
ERGONOMIC_OR_REPETITIVE_EXPOSURE	1	0.0605


Most Common Primary Causes
Caught In Or Between
Slip Trip Or Fall
Struck By Moving Object
Loss Of Control Of Tool Or Equipment
Awkward Movement Or Body Motion
Manual Handling Or Overexertion
Foreign Body Or Particle Contact
Heat Or Environmental Exposure
Other
Insufficient Information
Ground Or Roof Failure
Struck Against Stationary Object
0
50
100
150
200
250
300
350
Narrative count

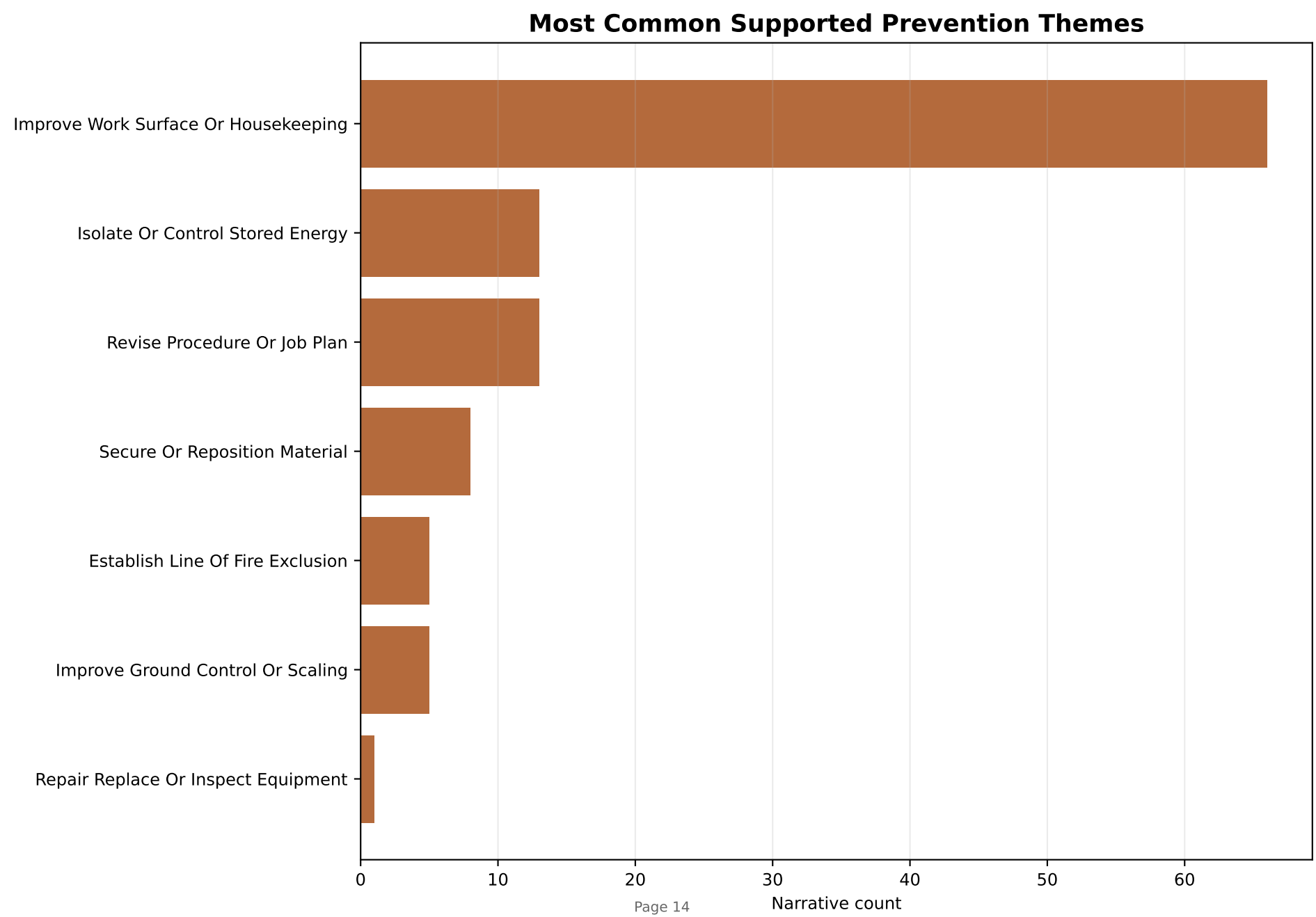
Supported Contributing-Factor Distribution

Contributing Factor	Count	Percent
POOR_HOUSEKEEPING_OR_SURFACE_CONDITION	66	59.4595
ENERGY_ISOLATION_DEFICIENCY	13	11.7117
PROCEDURE_OR_WORK_METHOD	11	9.9099
UNSTABLE_OR_IMPROPERLY_POSITIONED_MATER	8	7.2072
UNSTABLE_GROUND_OR_ROOF_CONDITION	5	4.5045
LINE_OF_FIRE_BODY_POSITION	5	4.5045
TRAINING_OR_EXPERIENCE	2	1.8018
EQUIPMENT_CONDITION_OR_FAILURE	1	0.9009


Most Common Supported Contributing Factors
Poor Housekeeping Or Surface Condition
Energy Isolation Deficiency
Procedure Or Work Method
Unstable Or Improperly Positioned Material
Line Of Fire Body Position
Unstable Ground Or Roof Condition
Training Or Experience
Equipment Condition Or Failure
0
10
20
30
40
50
60
Narrative count

Supported Prevention-Theme Distribution

Prevention Theme	Count	Percent
IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING	66	59.4595
REVISE_PROCEDURE_OR_JOB_PLAN	13	11.7117
ISOLATE_OR_CONTROL_STORED_ENERGY	13	11.7117
SECURE_OR_REPOSITION_MATERIAL	8	7.2072
IMPROVE_GROUND_CONTROL_OR_SCALING	5	4.5045
ESTABLISH_LINE_OF_FIRE_EXCLUSION	5	4.5045
REPAIR_REPLACE_OR_INSPECT_EQUIPMENT	1	0.9009


Most Common Supported Prevention Themes
Improve Work Surface Or Housekeeping
Isolate Or Control Stored Energy
Revise Procedure Or Job Plan
Secure Or Reposition Material
Establish Line Of Fire Exclusion
Improve Ground Control Or Scaling
Repair Replace Or Inspect Equipment
0
10
20
30
40
50
60
Narrative count

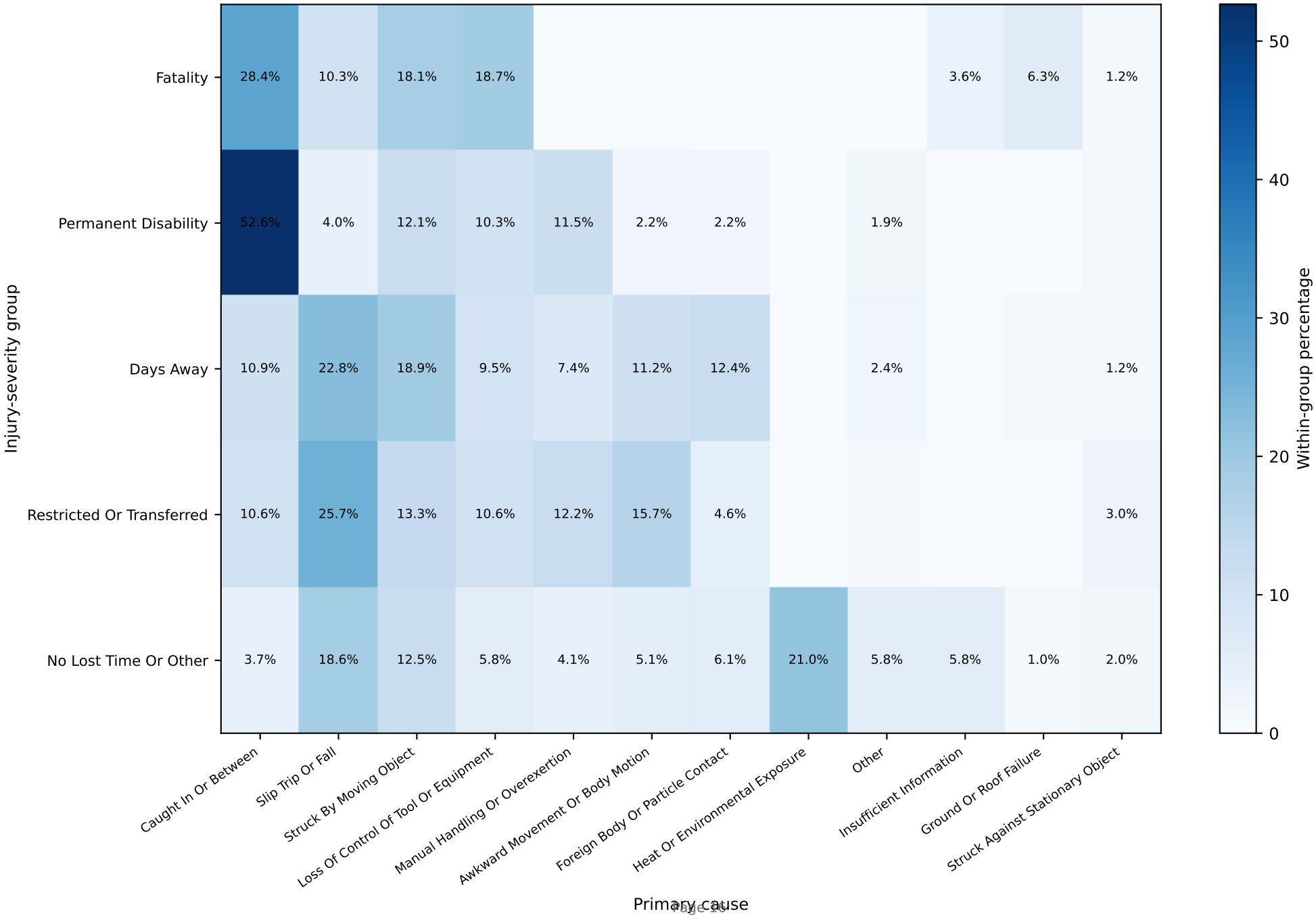
Severity Comparison

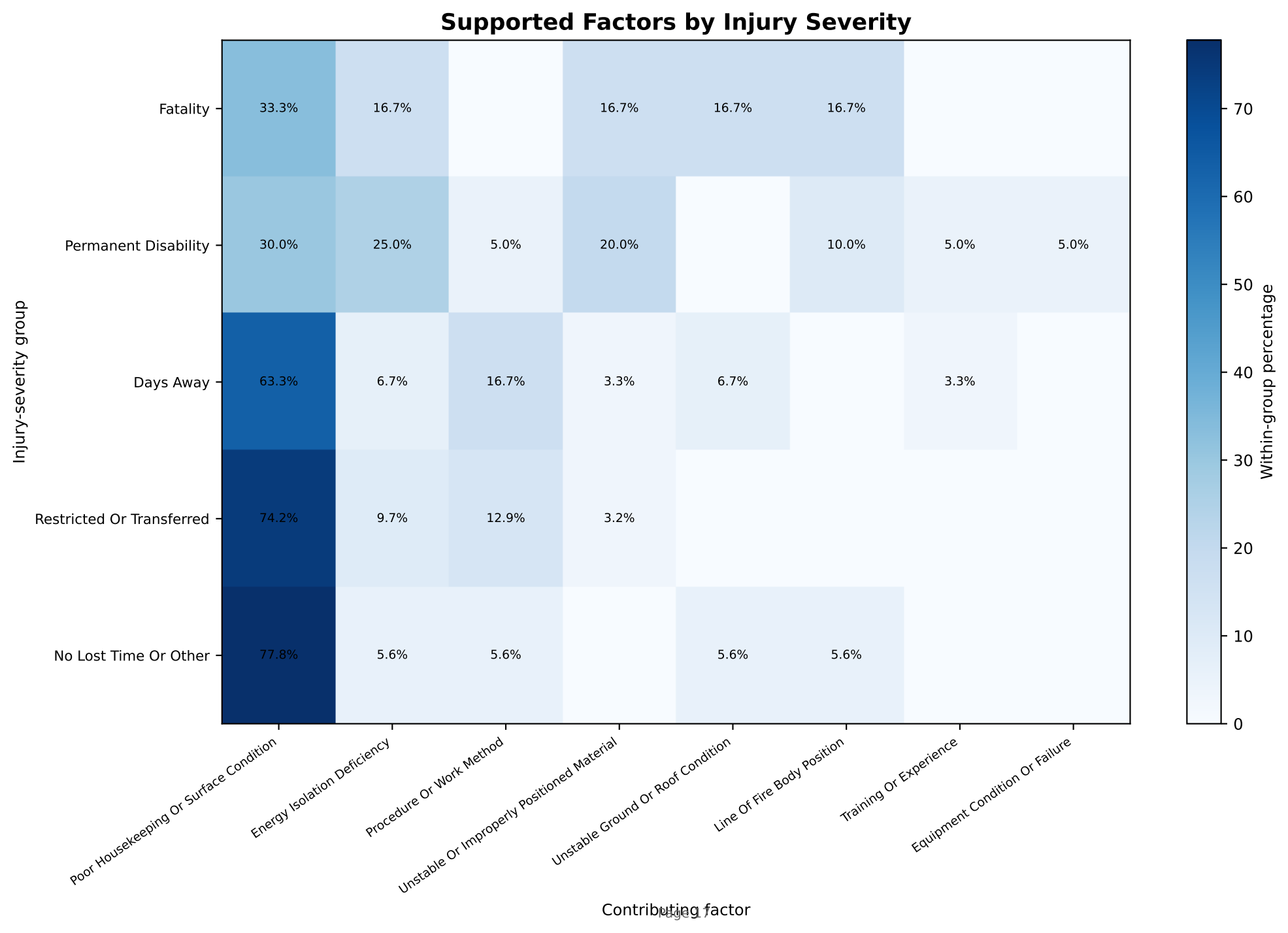
The following heatmaps show percentages within each severity group.

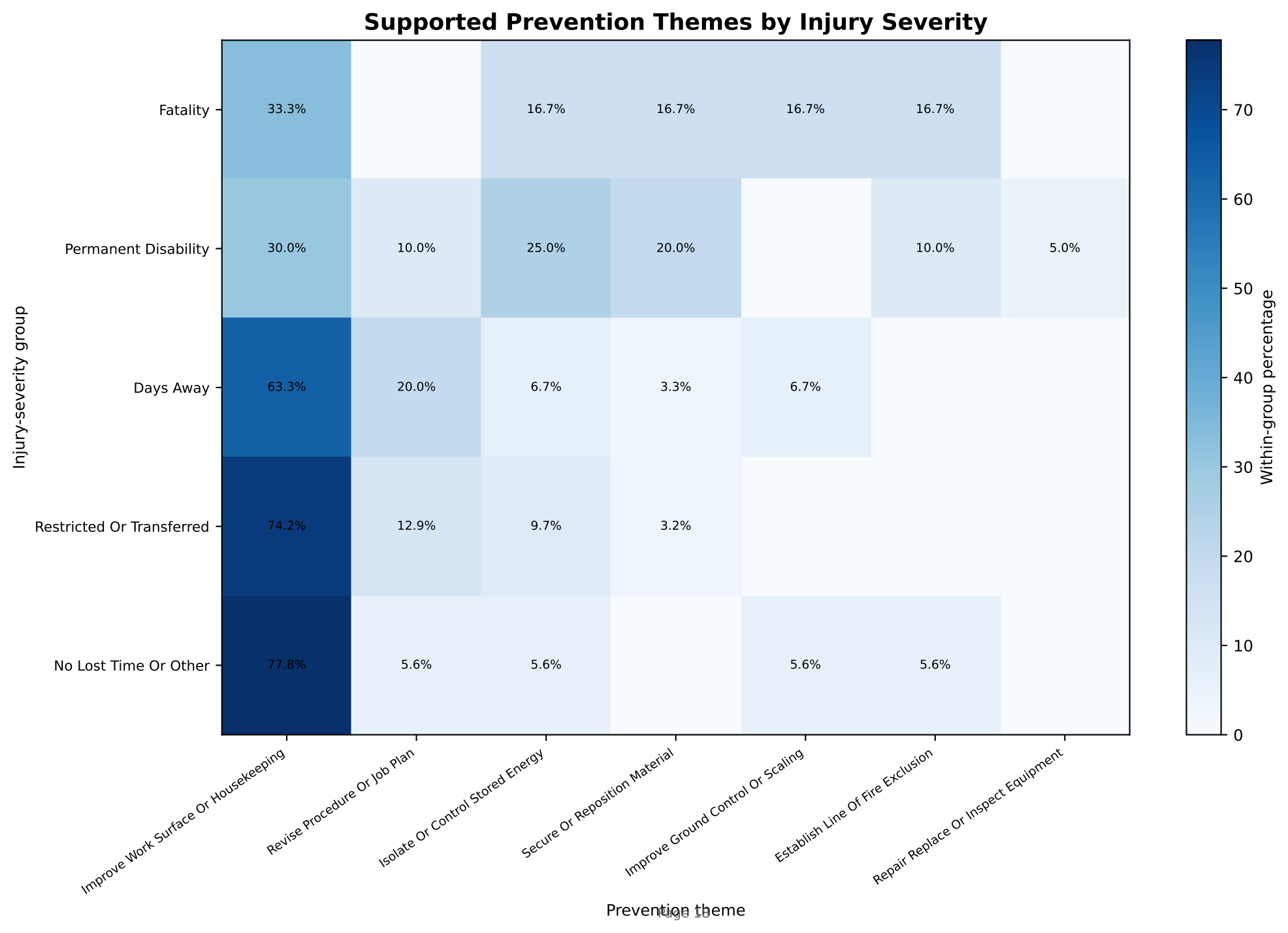
Cause results use the cause-analysis population.

Factor and prevention results use only the full-analysis population.

Observed differences are associations and do not establish that a specific cause or factor produces a particular severity.


Primary Causes by Injury Severity
Injury-severity group
Fatality
Permanent Disability
Days Away
Restricted Or Transferred
No Lost Time Or Other
28.4% 10.3% 18.1% 18.7% 3.6% 6.3% 1.2%
52.6% 4.0% 12.1% 10.3% 11.5% 2.2% 2.2% 1.9%
10.9% 22.8% 18.9% 9.5% 7.4% 11.2% 12.4% 2.4% 1.2%
10.6% 25.7% 13.3% 10.6% 12.2% 15.7% 4.6% 3.0%
3.7% 18.6% 12.5% 5.8% 4.1% 5.1% 6.1% 21.0% 5.8% 5.8% 1.0% 2.0%
Caught In Or Between
Slip Trip Or Fall
Struck By Moving Object
Loss Of Control Of Tool Or Equipment
Manual Handling Or Overexertion
Awkward Movement Or Body Motion
Foreign Body Or Particle Contact
Heat Or Environmental Exposure
Other
Insufficient Information
Ground Or Roof Failure
Struck Against Stationary Object
Primary cause
Within-group percentage
0
10
20
30
40
50


Supported Factors by Injury Severity
Injury-severity group
Fatality
Permanent Disability
Days Away
Restricted Or Transferred
No Lost Time Or Other
33.3%
16.7%
16.7%
16.7%
16.7%
30.0%
25.0%
5.0%
20.0%
10.0%
5.0%
5.0%
63.3%
6.7%
16.7%
3.3%
6.7%
3.3%
74.2%
9.7%
12.9%
3.2%
77.8%
5.6%
5.6%
5.6%
5.6%
Poor Housekeeping Or Surface Condition
Energy Isolation Deficiency
Procedure Or Work Method
Unstable Or Improperly Positioned Material
Unstable Ground Or Roof Condition
Line Of Fire Body Position
Training Or Experience
Equipment Condition Or Failure
Contributing factor
Within-group percentage
0
10
20
30
40
50
60
70


Supported Prevention Themes by Injury Severity
Injury-severity group
Fatality
Permanent Disability
Days Away
Restricted Or Transferred
No Lost Time Or Other
33.3%
16.7%
16.7%
16.7%
16.7%
30.0%
10.0%
25.0%
20.0%
10.0%
5.0%
63.3%
20.0%
6.7%
3.3%
6.7%
74.2%
12.9%
9.7%
3.2%
77.8%
5.6%
5.6%
5.6%
5.6%
Improve Work Surface Or Housekeeping
Revise Procedure Or Job Plan
Isolate Or Control Stored Energy
Secure Or Reposition Material
Improve Ground Control Or Scaling
Establish Line Of Fire Exclusion
Repair Replace Or Inspect Equipment
Prevention theme
Within-group percentage
0
10
20
30
40
50
60
70

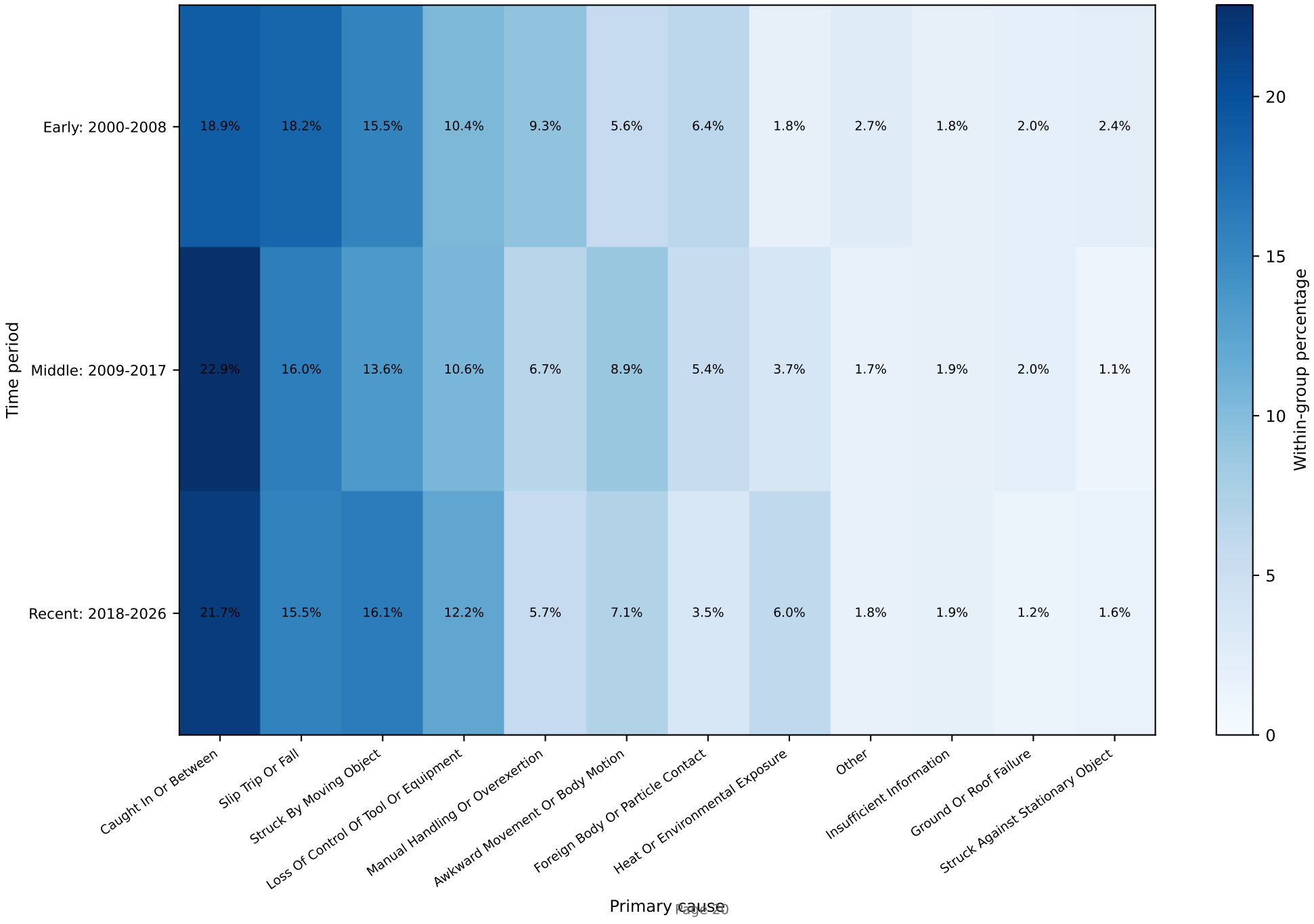
Time-Trend Interpretation

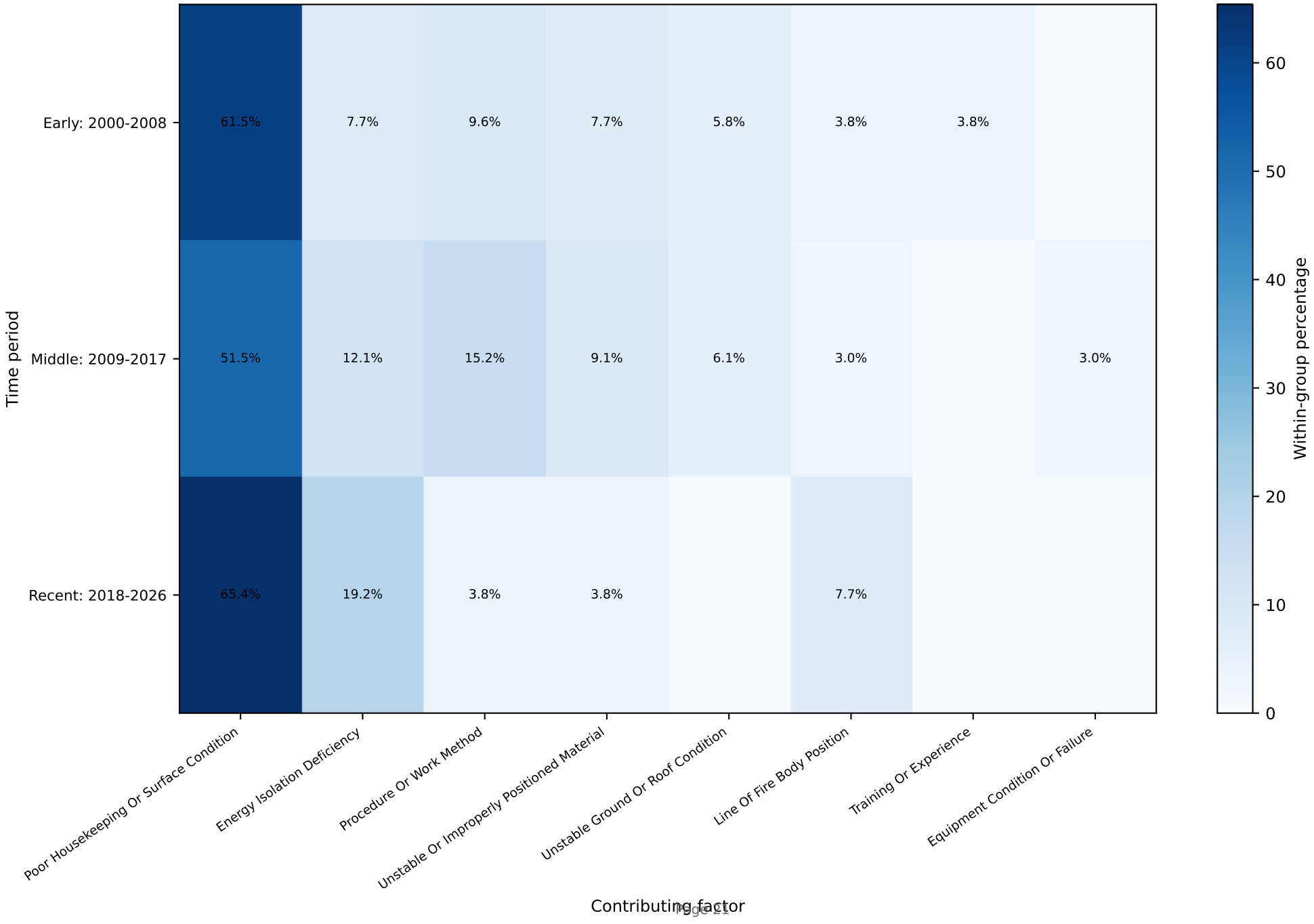
Time comparisons use within-period percentages.

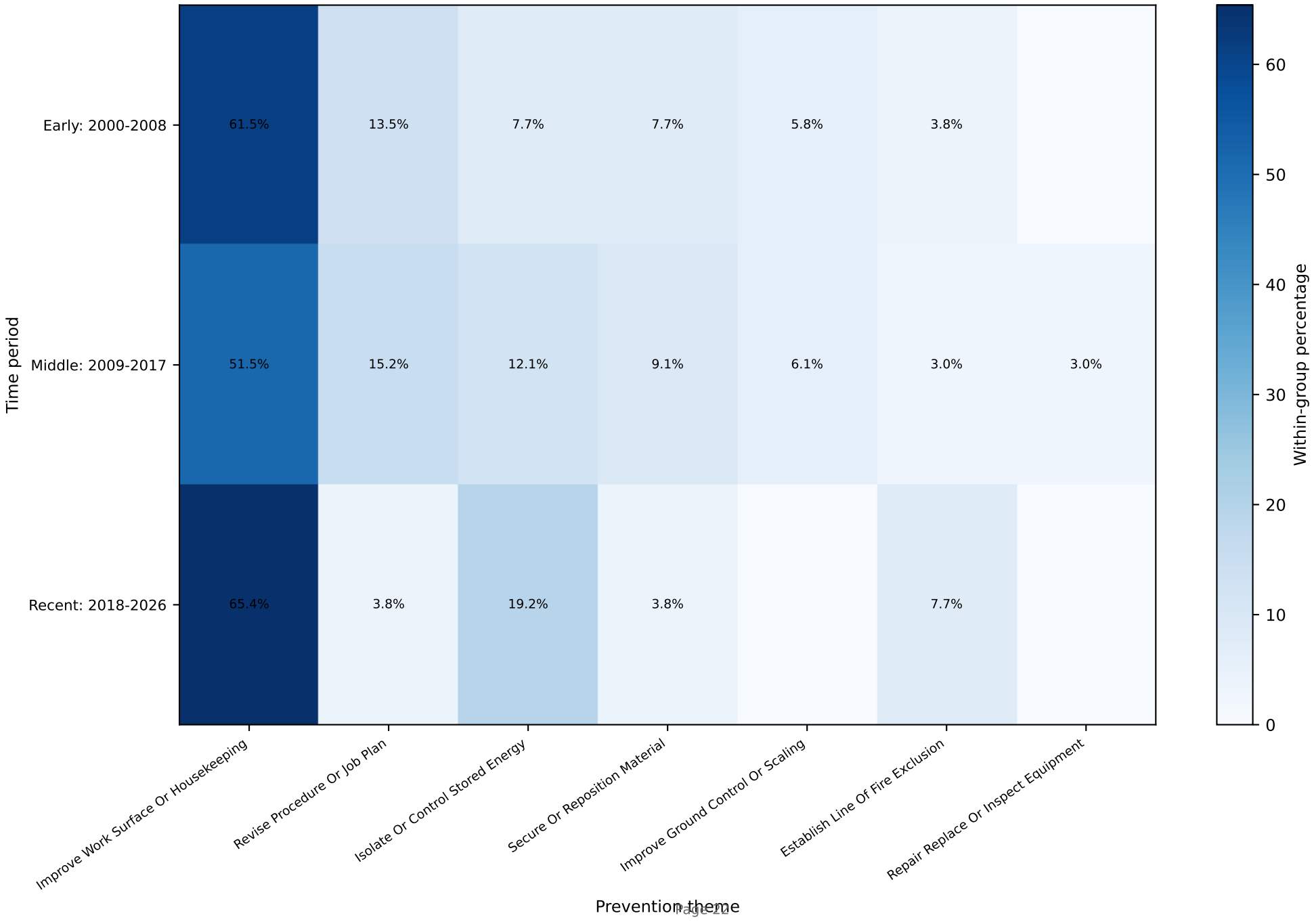
An increasing share means that a category represented a larger percentage of recent analyzed narratives than early narratives.

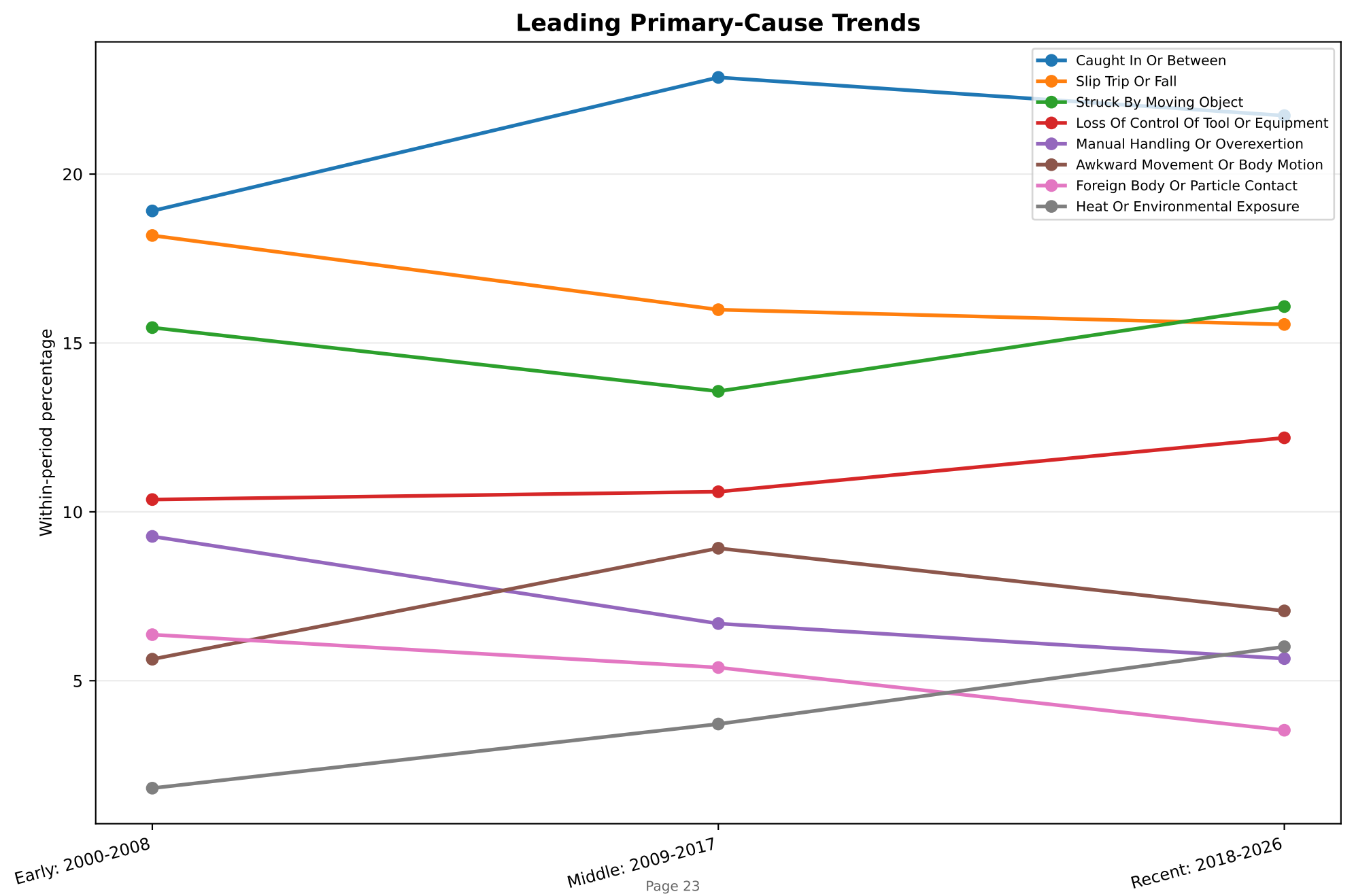
A declining share means that a category represented a smaller percentage.

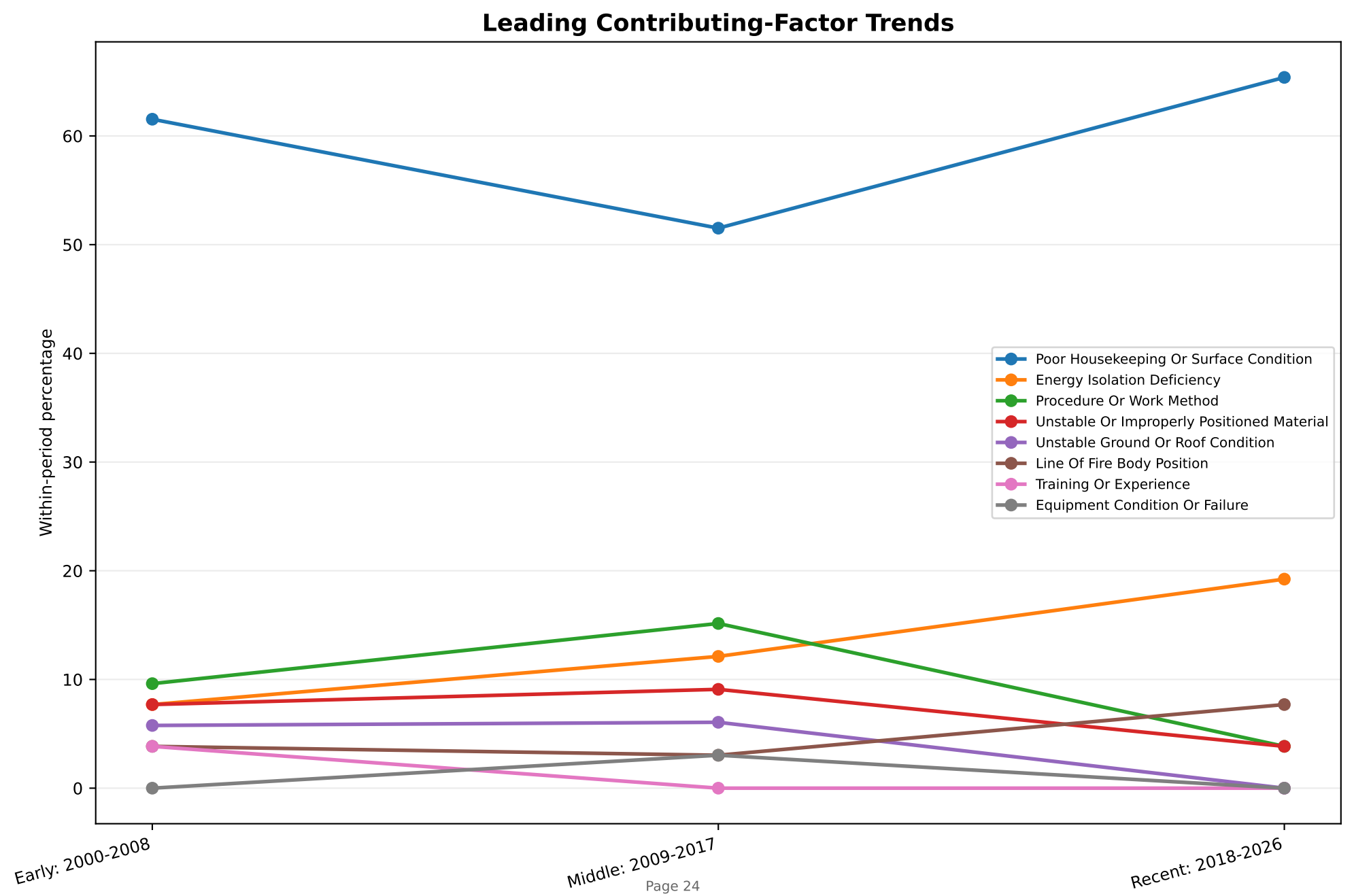
These results do not account for employee-hours, operating-mine count, production, contractor activity, sector composition, or reporting changes. They must not be described as exposure-adjusted incidence rates.

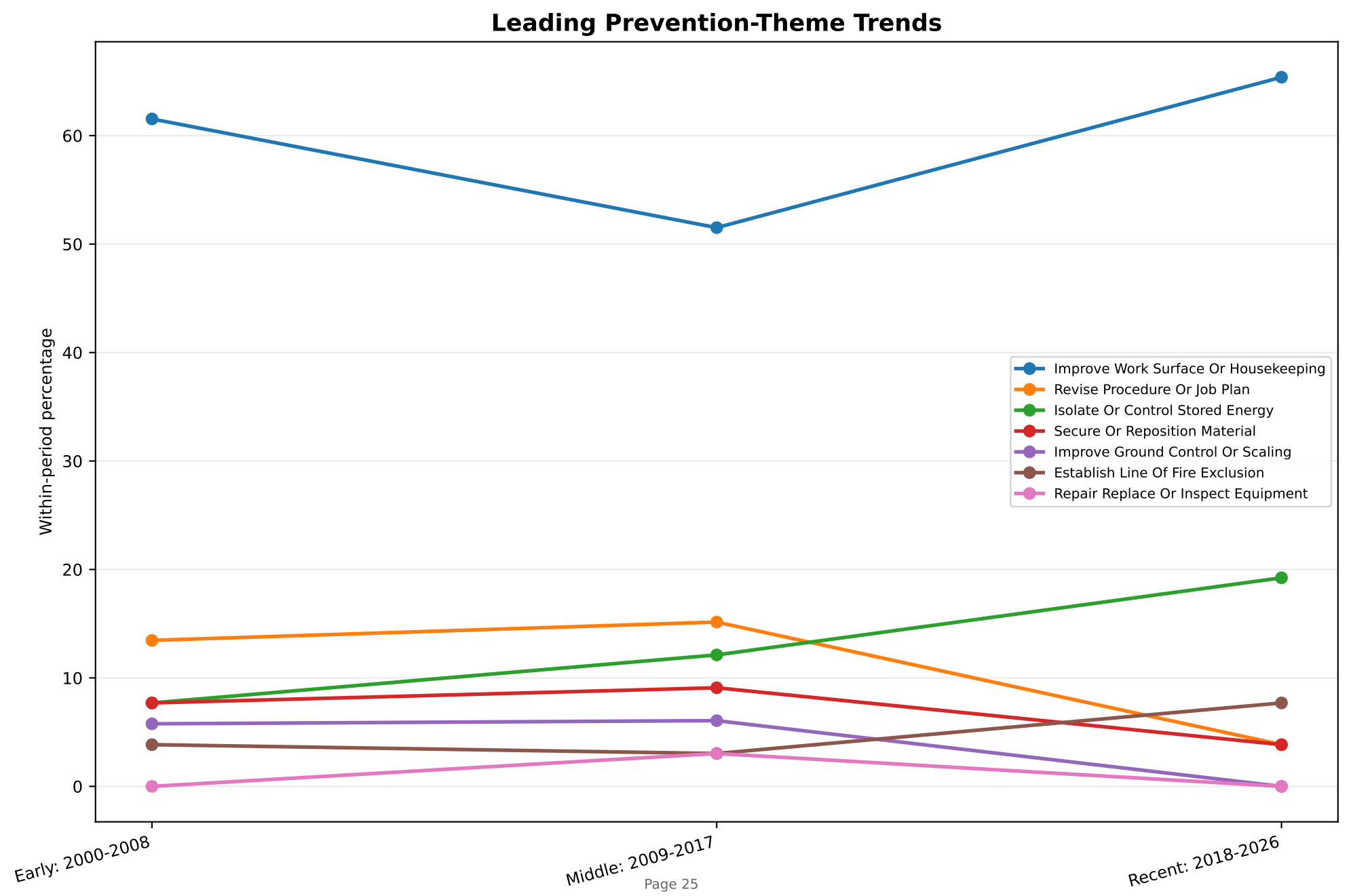

Primary Causes by Time Period
Time period
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
18.9% 18.2% 15.5% 10.4% 9.3% 5.6% 6.4% 1.8% 2.7% 1.8% 2.0% 2.4%
22.9% 16.0% 13.6% 10.6% 6.7% 8.9% 5.4% 3.7% 1.7% 1.9% 2.0% 1.1%
21.7% 15.5% 16.1% 12.2% 5.7% 7.1% 3.5% 6.0% 1.8% 1.9% 1.2% 1.6%
Caught In Or Between
Slip Trip Or Fall
Struck By Moving Object
Loss Of Control Of Tool Or Equipment
Manual Handling Or Overexertion
Awkward Movement Or Body Motion
Foreign Body Or Particle Contact
Heat Or Environmental Exposure
Other
Insufficient Information
Ground Or Roof Failure
Struck Against Stationary Object
Primary cause
Within-group percentage
0
5
10
15
20


Supported Factors by Time Period
Time period
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
61.5%
7.7%
9.6%
7.7%
5.8%
3.8%
3.8%
51.5%
12.1%
15.2%
9.1%
6.1%
3.0%
3.0%
65.4%
19.2%
3.8%
3.8%
7.7%
Poor Housekeeping Or Surface Condition
Energy Isolation Deficiency
Procedure Or Work Method
Unstable Or Improperly Positioned Material
Unstable Ground Or Roof Condition
Line Of Fire Body Position
Training Or Experience
Equipment Condition Or Failure
Contributing factor
Within-group percentage
0
10
20
30
40
50
60


Supported Prevention Themes by Time Period
Time period
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
61.5%
13.5%
7.7%
7.7%
5.8%
3.8%
51.5%
15.2%
12.1%
9.1%
6.1%
3.0%
3.0%
65.4%
3.8%
19.2%
3.8%
7.7%
Improve Work Surface Or Housekeeping
Revise Procedure Or Job Plan
Isolate Or Control Stored Energy
Secure Or Reposition Material
Improve Ground Control Or Scaling
Establish Line Of Fire Exclusion
Repair Replace Or Inspect Equipment
Prevention theme
Within-group percentage
0
10
20
30
40
50
60


Leading Primary-Cause Trends
Caught In Or Between
Slip Trip Or Fall
Struck By Moving Object
Loss Of Control Of Tool Or Equipment
Manual Handling Or Overexertion
Awkward Movement Or Body Motion
Foreign Body Or Particle Contact
Heat Or Environmental Exposure
Within-period percentage
20
15
10
5
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026


Leading Contributing-Factor Trends
Within-period percentage
0
10
20
30
40
50
60
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
Poor Housekeeping Or Surface Condition
Energy Isolation Deficiency
Procedure Or Work Method
Unstable Or Improperly Positioned Material
Unstable Ground Or Roof Condition
Line Of Fire Body Position
Training Or Experience
Equipment Condition Or Failure


Leading Prevention-Theme Trends
Within-period percentage
0
10
20
30
40
50
60
Early: 2000-2008
Middle: 2009-2017
Recent: 2018-2026
Improve Work Surface Or Housekeeping
Revise Procedure Or Job Plan
Isolate Or Control Stored Energy
Secure Or Reposition Material
Improve Ground Control Or Scaling
Establish Line Of Fire Exclusion
Repair Replace Or Inspect Equipment

Early-to-Recent Cause Changes

Primary Cause	Total Count	Early Percent	Recent Percent	Percentage-Point Change	Direction
HEAT_OR_ENVIRONMENTAL_EXP(	64	1.8182	6.0071	4.1889	Increasing share
CAUGHT_IN_OR_BETWEEN	350	18.9091	21.7314	2.8224	Increasing share
LOSS_OF_CONTROL_OF_TOOL_O	183	10.3636	12.1908	1.8272	Increasing share
AWKWARD_MOVEMENT_OR_BOD	119	5.6364	7.0671	1.4308	Increasing share
CHEMICAL_DUST_OR_OCCUPATIC	28	1.4545	2.2968	0.8423	Approximately stable
STRUCK_BY_MOVING_OBJECT	249	15.4545	16.0777	0.6232	Approximately stable
PRESSURE_OR_STORED_ENERGY	21	0.7273	1.0601	0.3328	Approximately stable
INSUFFICIENT_INFORMATION	31	1.8182	1.9435	0.1253	Approximately stable
UNEXPECTED_EQUIPMENT_MOVE	12	0.7273	0.7067	-0.0206	Approximately stable
GROUND_OR_ROOF_FAILURE	29	2.0000	1.2367	-0.7633	Approximately stable
STRUCK_AGAINST_STATIONARY_(	28	2.3636	1.5901	-0.7735	Approximately stable
ELECTRICAL_CONTACT	18	1.6364	0.7067	-0.9296	Approximately stable
OTHER	34	2.7273	1.7668	-0.9605	Approximately stable
SLIP_TRIP_OR_FALL	274	18.1818	15.5477	-2.6341	Declining share
FOREIGN_BODY_OR_PARTICLE_C(	84	6.3636	3.5336	-2.8301	Declining share
MANUAL_HANDLING_OR_OVEREX	119	9.2727	5.6537	-3.6190	Declining share

Early-to-Recent Factor Changes

Contributing Factor	Total Count	Early Percent	Recent Percent	Percentage-Point Change	Direction
ENERGY_ISOLATION_DEFICIENCY	13	7.6923	19.2308	11.5385	Increasing share
POOR_HOUSEKEEPING_OR_SURF	66	61.5385	65.3846	3.8462	Increasing share
PROCEDURE_OR_WORK_METHOI	11	9.6154	3.8462	-5.7692	Declining share

Early-to-Recent Prevention Changes

Prevention Theme	Total Count	Early Percent	Recent Percent	Percentage-Point Change	Direction
ISOLATE_OR_CONTROL_STORED_	13	7.6923	19.2308	11.5385	Increasing share
IMPROVE_WORK_SURFACE_OR_H	66	61.5385	65.3846	3.8462	Increasing share
REVISE_PROCEDURE_OR_JOB_PL	13	13.4615	3.8462	-9.6154	Declining share

Statistical Comparison Method

Chi-square tests compare category distributions among severity groups and time periods.

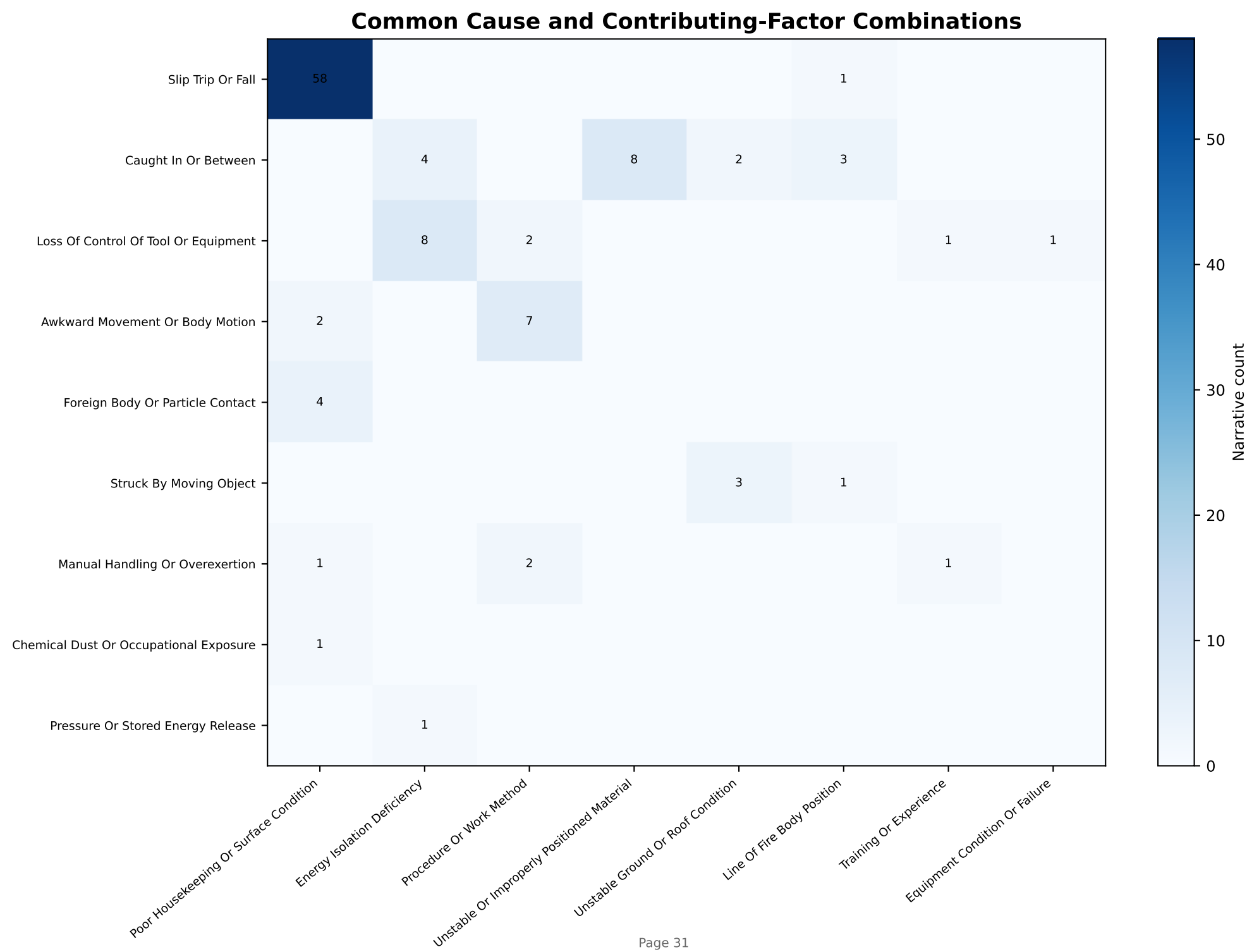
Cramer's V summarizes association strength from zero to one.

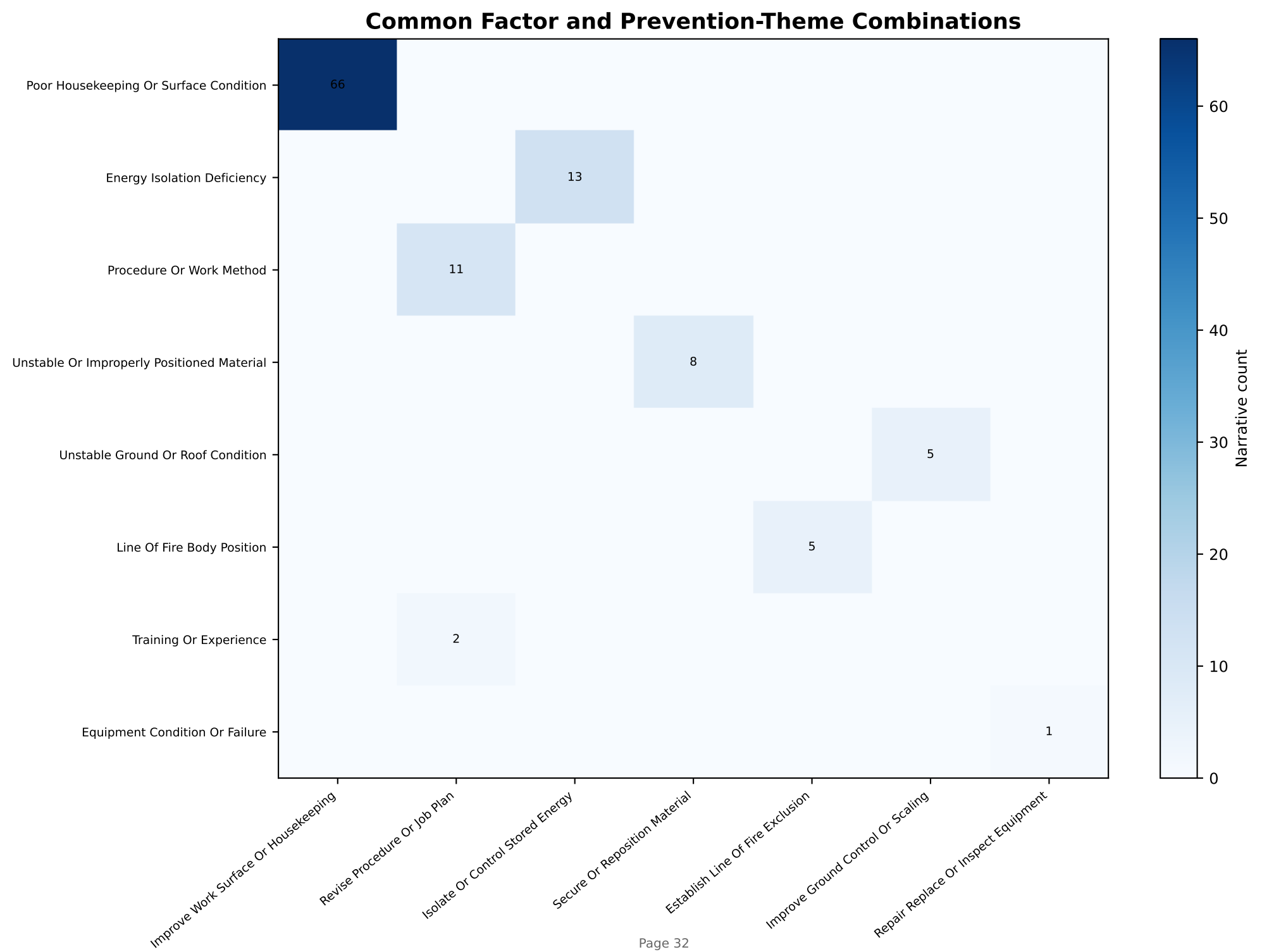
Statistical significance should be interpreted with category counts, percentage differences, effect size, and practical importance.

These tests do not establish causation and do not adjust for exposure or changing industry composition.

Chi-Square and Cramer's V Results

Comparison	Chi-Square	Degrees Of Freedom	P-Value	Cramer'S V	Records
Primary cause by severity	1079.7737	72	0.0000	0.4040	1654
Contributing factor by severity	45.2903	28	0.0206	0.3194	111
Prevention theme by severity	42.8180	24	0.0104	0.3105	111
Primary cause by time period	51.8343	36	0.0425	0.1252	1654
Contributing factor by time peri	11.9517	14	0.6102	0.2320	111
Prevention theme by time perio	9.6412	12	0.6474	0.2084	111


Common Cause and Contributing-Factor Combinations
Slip Trip Or Fall
Caught In Or Between
Loss Of Control Of Tool Or Equipment
Awkward Movement Or Body Motion
Foreign Body Or Particle Contact
Struck By Moving Object
Manual Handling Or Overexertion
Chemical Dust Or Occupational Exposure
Pressure Or Stored Energy Release
Poor Housekeeping Or Surface Condition
Energy Isolation Deficiency
Procedure Or Work Method
Unstable Or Improperly Positioned Material
Unstable Ground Or Roof Condition
Line Of Fire Body Position
Training Or Experience
Equipment Condition Or Failure
Narrative count
0
10
20
30
40
50


Common Factor and Prevention-Theme Combinations
Poor Housekeeping Or Surface Condition
Energy Isolation Deficiency
Procedure Or Work Method
Unstable Or Improperly Positioned Material
Unstable Ground Or Roof Condition
Line Of Fire Body Position
Training Or Experience
Equipment Condition Or Failure
66
13
11
8
5
5
2
1
Improve Work Surface Or Housekeeping
Revise Procedure Or Job Plan
Isolate Or Control Stored Energy
Secure Or Reposition Material
Establish Line Of Fire Exclusion
Improve Ground Control Or Scaling
Repair Replace Or Inspect Equipment
Narrative count
0
10
20
30
40
50
60

Quality-Sensitivity Analysis

Cause distributions are compared across three populations:

1. All successfully processed acute-accident records.

2. Cause-valid records, including conservative no-factor cases.

3. Full-analysis records with supported factors and aligned prevention themes.

The maximum percentage spread shows how much a cause's estimated share changes under stricter quality filters.

Cause Sensitivity to Quality Filters

Primary Cause	All Successful Acute	Cause-Valid Population	Full-Analysis Population	Maximum Percentage Spread
SLIP_TRIP_OR_FALL	15.3172	16.5659	53.1532	37.8360
STRUCK_BY_MOVING_OBJECT	15.3172	15.0544	3.6036	11.7136
CAUGHT_IN_OR_BETWEEN	22.8984	21.1608	15.3153	7.5831
HEAT_OR_ENVIRONMENTAL_EXPOSURE	3.4554	3.8694	0.0000	3.8694
MANUAL_HANDLING_OR_OVEREXERTIOI	6.4466	7.1947	3.6036	3.5911
INSUFFICIENT_INFORMATION	3.2491	1.8742	0.0000	3.2491
OTHER	1.9598	2.0556	0.0000	2.0556
GROUND_OR_ROOF_FAILURE	1.7019	1.7533	0.0000	1.7533
STRUCK_AGAINST_STATIONARY_OBJECT	1.6503	1.6929	0.0000	1.6929
FOREIGN_BODY_OR_PARTICLE_CONTACT	4.6416	5.0786	3.6036	1.4750
AWKWARD_MOVEMENT_OR_BODY_MOTI	6.7561	7.1947	8.1081	1.3520
ELECTRICAL_CONTACT	0.9799	1.0883	0.0000	1.0883
CHEMICAL_DUST_OR_OCCUPATIONAL_E	1.5988	1.6929	0.9009	0.7920
UNEXPECTED_EQUIPMENT_MOVEMENT	0.6189	0.7255	0.0000	0.7255
VEHICLE_OR_MOBILE_EQUIPMENT_EVEN	0.5157	0.3628	0.0000	0.5157
LOSS_OF_CONTROL_OF_TOOL_OR_EQUI	11.2945	11.0641	10.8108	0.4837
PRESSURE_OR_STORED_ENERGY_RELEA	1.2893	1.2696	0.9009	0.3884
FIRE_EXPLOSION_OR_THERMAL_EVENT	0.2579	0.2418	0.0000	0.2579
ERGONOMIC_OR_REPETITIVE_EXPOSUR	0.0516	0.0605	0.0000	0.0605

Representative Fully Validated Records - 1 of 4

These records met the full-analysis requirements with valid cause evidence, supported factor evidence, aligned prevention, and no critical review condition.

==========
Example 1 | Record 220042610058 | Days away | Early: 2000-2008
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: CAUGHT_IN_OR_BETWEEN
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Prevention: SECURE_OR_REPOSITION_MATERIAL
Critical review:
Advisory review:

Narrative:
EE WAS EXITING MINE AT END OF SHIFT. EE'S PICKUP TRUCK STALLED (QUIT) COMING UP SLOPE. SERVICE TRUCK WAS SUMMONED TO TOW PICKUP. EE WAS ASSISTING HOOKING SERVICE TRUCK TO PICKUP. THE SERVICE TRUCK SLID IN GRAVEL AND PINNED EE'S RIGHT LEG BETWEEN SERVICE TRUCK AND PICKUP.

Cause evidence: THE SERVICE TRUCK SLID IN GRAVEL AND PINNED EE'S RIGHT LEG BETWEEN SERVICE TRUCK AND PICKUP.
Factor evidence: THE SERVICE TRUCK SLID IN GRAVEL AND PINNED EE'S RIGHT LEG BETWEEN SERVICE TRUCK AND PICKUP.
Prevention rationale: Secure or reposition the service truck and pickup to prevent them from sliding and creating a caught-in-between hazard.

==========
Example 2 | Record 220151810023 | Permanent disability | Middle: 2009-2017
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: CAUGHT_IN_OR_BETWEEN
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Prevention: SECURE_OR_REPOSITION_MATERIAL
Critical review:
Advisory review:

Narrative:
In process of switching rail cars. Employee attempted to kick knuckles of rail car to line up for connection. Foot slipped pinning foot between the two knuckles as cars were pushed together. Broken bones / lacerations.

Cause evidence: Employee attempted to kick knuckles of rail car to line up for connection. Foot slipped pinning foot between the two knuckles as cars were pushed together.
Factor evidence: Foot slipped pinning foot between the two knuckles of the rail car.
Prevention rationale: To prevent this, the knuckles should be secured or repositioned to avoid the risk of the foot getting caught.

==========
Example 3 | Record 220090630008 | Restricted or transferred | Middle: 2009-2017
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: SLIP_TRIP_OR_FALL

Representative Fully Validated Records - 2 of 4

Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Critical review:
Advisory review:

Narrative:
slipped off of step of dozer and twisted ankle.

Cause evidence: slipped off of step of dozer
Factor evidence: twisted ankle
Prevention rationale: Improving the work surface condition by ensuring steps are secure and free of hazards can prevent slips.

==
Example 4 | Record 220240800009 | Days away | Recent: 2018-2026
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: SLIP_TRIP_OR_FALL
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Critical review:
Advisory review:

Narrative:
IE was walking down the slope when IE slipped and fell. The employee hit knee causing a bruise.

Cause evidence: IE was walking down the slope when IE slipped and fell.
Factor evidence: The employee hit their knee causing a bruise.
Prevention rationale: Improving the work surface condition by addressing any hazards such as loose rocks, uneven footing, or other surface conditions can prevent slips and falls.

==
Example 5 | Record 220193180005 | Restricted or transferred | Recent: 2018-2026
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: SLIP_TRIP_OR_FALL
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Critical review:
Advisory review:

Narrative:
miner tripped and fell injuring EE's right shoulder, right shoulder contusion injury

Cause evidence: miner tripped and fell injuring EE's right shoulder, right shoulder contusion injury
Factor evidence: miner tripped and fell injuring EE's right shoulder, right shoulder contusion injury
Prevention rationale: Improving the work surface condition can reduce the risk of tripping and falling.

==
Example 6 | Record 220031630010 | Fatality | Early: 2000-2008

Representative Fully Validated Records - 3 of 4

Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: CAUGHT_IN_OR_BETWEEN
Factor: UNSTABLE_GROUND_OR_ROOF_CONDITION
Prevention: IMPROVE_GROUND_CONTROL_OR_SCALING
Critical review:
Advisory review:

Narrative:
EMBANKMENT CAVE IN RESULTED IN PINNING VICTIM IN FROM CHEST DOWN THUS CONSTRICTING BREATHING WHICH ACCORDING TO CORONER RESULTED IN ASPHYXIATION.

Cause evidence: EMBANKMENT CAVE IN RESULTED IN PINNING VICTIM IN FROM CHEST DOWN THUS CONSTRICTING BREATHING.
Factor evidence: EMBANKMENT CAVE IN
Prevention rationale: Improving ground control and scaling would help prevent embankment cave-ins and reduce the risk of victims being caught between the ground and the embankment.

==
Example 7 | Record 220022800112 | Days away | Early: 2000-2008
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: SLIP_TRIP_OR_FALL
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Critical review:
Advisory review:

Narrative:
CLIMBING UP IN TRUCK. HE SLIPPED OFF AND FELL.

Cause evidence: HE SLIPPED OFF AND FELL.
Factor evidence: HE SLIPPED OFF AND FELL.
Prevention rationale: Improving the work surface condition can prevent slips and falls.

==
Example 8 | Record 220001510001 | Fatality | Early: 2000-2008
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: SLIP_TRIP_OR_FALL
Factor: POOR_HOUSEKEEPING_OR_SURFACE_CONDITION
Prevention: IMPROVE_WORK_SURFACE_OR_HOUSEKEEPING
Critical review:
Advisory review:

Narrative:
WAS NOT WEARING LIFT JACKET WHEN CLEANING OUT DEBRIS FROM CUTTERHEAD, FELL INTO WATER AND DROWNED.

Cause evidence: WAS NOT WEARING LIFT JACKET WHEN CLEANING OUT DEBRIS FROM CUTTERHEAD, FELL INTO WATER AND DROWNED.
Factor evidence: FELL INTO WATER AND DROWNED.

Representative Fully Validated Records - 4 of 4

Prevention rationale: Improving work surface conditions and ensuring proper protective equipment can prevent falls into water hazards.

==
Example 9 | Record 220142720035 | Permanent disability | Middle: 2009-2017
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: CAUGHT_IN_OR_BETWEEN
Factor: ENERGY_ISOLATION_DEFICIENCY
Prevention: ISOLATE_OR_CONTROL_STORED_ENERGY
Critical review:
Advisory review:

Narrative:
Maintenance employee failed to follow LOTO procedure while working on pinch tube mechanism. Pinch tube activated and crushed employee's left thumb causing loss of thumb tip at first joint.

Cause evidence: Pinch tube activated and crushed employee's left thumb causing loss of thumb tip at first joint.
Factor evidence: Maintenance employee failed to follow LOTO procedure while working on pinch tube mechanism.
Prevention rationale: Implement and enforce lockout/tagout (LOTO) procedures to ensure stored energy is properly isolated before performing maintenance.

==
Example 10 | Record 220180680018 | Days away | Recent: 2018-2026
Analysis status: VALID_FOR_FULL_ANALYSIS
Cause: PRESSURE_OR_STORED_ENERGY_RELEASE
Factor: ENERGY_ISOLATION_DEFICIENCY
Prevention: ISOLATE_OR_CONTROL_STORED_ENERGY
Critical review:
Advisory review:

Narrative:
Another tech and myself started to disassemble the accumulator and failed to bleed off the nitrogen before starting disassembly. I was holding the ratchet on the end cap while ee ran the ratchet. When the end cap came to the end of its threads the pressure that was still in the accumulator blew the end cap into my hand.

Cause evidence: When the end cap came to the end of its threads the pressure that was still in the accumulator blew the end cap into my hand.
Factor evidence: Failed to bleed off the nitrogen before starting disassembly.
Prevention rationale: Ensure all stored energy is released before beginning disassembly to prevent accidental release of pressure.

Cause-Valid Records with No Separate Factor - 1 of 5

These narratives support the event mechanism but do not separately support why the event occurred.

==
Example 1 | Record 220140630085 | Permanent disability | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
Employees were attempting to install a 8"X8"X1" steel diagonal brace. The upper end of the brace slipped off the connection clip on the support column and struck the right hand of Employee #1 pinching it between the diagonal brace and the inside of the support column. Physicians at UMC-Tucson made the decision to amputate the right pinky finger.

Cause evidence: The upper end of the brace slipped off the connection clip on the support column and struck the right hand of Employee #1 pinching it between the diagonal brace and the inside of the support column.
Factor evidence:
Prevention rationale: Repair or replace the connection clip on the support column to ensure the brace remains securely in place during installation.

==
Example 2 | Record 220262430047 | No lost time or other | Recent: 2018-2026
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: HEAT_OR_ENVIRONMENTAL_EXPOSURE
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
Employee was shoveling material off of a conveyor belt. The temp outside was 103 during this time. the employee experienced some cramping. Employee was brought to the office to cool off and hydrate but did not feel any improvement. They were checked out at the hospital and received an IV and potassium.

Cause evidence: The temperature outside was 103 during the time the employee was shoveling material off of a conveyor belt, and the employee experienced cramping.
Factor evidence:
Prevention rationale: Implement cooling stations and hydration breaks to prevent heat-related illnesses in hot environments.

==
Example 3 | Record 220082740067 | Days away | Early: 2000-2008

Cause-Valid Records with No Separate Factor - 2 of 5

Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: ELECTRICAL_CONTACT
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Injured was handling roof drill cable when he came in contact with lead causing burn to hand.

Cause evidence: Injured was handling roof drill cable when he came in contact with lead causing burn to hand.
Factor evidence:
Prevention rationale: To prevent electrical contact injuries, the stored electrical energy should be isolated or controlled.

==========
Example 4 | Record 220241500023 | No lost time or other | Recent: 2018-2026
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: HEAT_OR_ENVIRONMENTAL_EXPOSURE
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
An excavator operator became exhausted by heat when the air conditioning was not working properly in the cab. The operator was told several times during the shift by supervision that EE could park the machine if it became too hot to operate but failed to do so.

Cause evidence: The excavator operator became exhausted by heat when the air conditioning was not working properly in the cab.
Factor evidence:
Prevention rationale: Ensure the air conditioning system in the excavator cab is functioning properly to control the environmental conditions and prevent heat-related injuries.

==========
Example 5 | Record 220091180011 | Fatality | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Cause-Valid Records with No Separate Factor - 3 of 5

Narrative:
Employee was pinned between concrete structure and wall of hole.

Cause evidence: Employee was pinned between concrete structure and wall of hole.
Factor evidence:
Prevention rationale: Securing or repositioning the concrete structure and wall of the hole would prevent the employee from being caught in between them.

==
Example 6 | Record 220063350035 | Days away | Early: 2000-2008
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: STRUCK_BY_MOVING_OBJECT
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
Employee was operating miner when a piece of rock fell hitting him on the neck. Rock measured 4'6" long, 2' wide and 2" thick. Doctor placed him on medical leave effective 11/27/2006.

Cause evidence: A piece of rock fell and hit the employee on the neck.
Factor evidence:
Prevention rationale: To prevent similar incidents, the rock should be properly secured or removed to avoid falling and injuring workers.

==
Example 7 | Record 220140920009 | Fatality | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
A miner become buried from material sliding down the face of a bank.

Cause evidence: A miner became buried from material sliding down the face of a bank.
Factor evidence:
Prevention rationale: To prevent future incidents, the material should be stabilized or the movement of the material should be controlled.

==
Example 8 | Record 220100710025 | Days away | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY

Cause-Valid Records with No Separate Factor - 4 of 5

Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
EE was in process of changing out troughing idler on 30" conv. belt. While performing this task with a co-worker the idler tilted down and came in contact with the conv. frame. EE's right middle finger was mashed between the idler and the conv. frame. EE's right middle finger,last joint recieved approx. 11 stitches.

Cause evidence: EE's right middle finger was mashed between the idler and the conveyor frame while changing out the troughing idler.
Factor evidence:
Prevention rationale: Repair or replace the idler to ensure it remains stable during maintenance tasks and inspect all equipment before performing maintenance to prevent similar incidents.

==========
Example 9 | Record 220192070010 | Restricted or transferred | Recent: 2018-2026
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: MANUAL_HANDLING_OR_OVEREXERTION
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Employee was attempting to pick up a 50lb bag of floor dry and felt pain in their left wrist and shoulder.

Cause evidence: Employee was attempting to pick up a 50lb bag of floor dry and felt pain in their left wrist and shoulder.
Factor evidence:
Prevention rationale: Provide targeted training on proper lifting techniques and ergonomics to reduce the risk of strain and injury during manual handling tasks.

==========
Example 10 | Record 220050970011 | Days away | Early: 2000-2008
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Cause-Valid Records with No Separate Factor - 5 of 5

Narrative:
Employee was coming around the canopy (making a turn to the right) when his finger tips were caught in the swivel point. This resulted in two finger tips being pinched & crushed.

Cause evidence: Employee was coming around the canopy (making a turn to the right) when his finger tips were caught in the swivel point. This resulted in two finger tips being pinched & crushed.
Factor evidence:
Prevention rationale: Implement mechanical or physical controls to isolate the swivel point from the employee's reach to prevent catching and pinching injuries.

Unsupported-Factor Safeguard Examples - 1 of 5

The original factor lacked supporting narrative evidence. The factor was conservatively replaced with NO_SEPARATE_FACTOR_STATED and the prevention theme was changed to INVESTIGATION_REQUIRED.

==
Example 1 | Record 220140630085 | Permanent disability | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
Employees were attempting to install a 8"X8"X1" steel diagonal brace. The upper end of the brace slipped off the connection clip on the support column and struck the right hand of Employee #1 pinching it between the diagonal brace and the inside of the support column. Physicians at UMC-Tucson made the decision to amputate the right pinky finger.

Cause evidence: The upper end of the brace slipped off the connection clip on the support column and struck the right hand of Employee #1 pinching it between the diagonal brace and the inside of the support column.
Factor evidence:
Prevention rationale: Repair or replace the connection clip on the support column to ensure the brace remains securely in place during installation.

==
Example 2 | Record 220262430047 | No lost time or other | Recent: 2018-2026
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: HEAT_OR_ENVIRONMENTAL_EXPOSURE
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
Employee was shoveling material off of a conveyor belt. The temp outside was 103 during this time. the employee experienced some cramping. Employee was brought to the office to cool off and hydrate but did not feel any improvement. They were checked out at the hospital and received an IV and potassium.

Cause evidence: The temperature outside was 103 during the time the employee was shoveling material off of a conveyor belt, and the employee experienced cramping.
Factor evidence:
Prevention rationale: Implement cooling stations and hydration breaks to prevent heat-related illnesses in hot environments.

==

Unsupported-Factor Safeguard Examples - 2 of 5

Example 3 | Record 220241500023 | No lost time or other | Recent: 2018-2026
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: HEAT_OR_ENVIRONMENTAL_EXPOSURE
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
An excavator operator became exhausted by heat when the air conditioning was not working properly in the cab. The operator was told several times during the shift by supervision that EE could park the machine if it became too hot to operate but failed to do so.

Cause evidence: The excavator operator became exhausted by heat when the air conditioning was not working properly in the cab.
Factor evidence:
Prevention rationale: Ensure the air conditioning system in the excavator cab is functioning properly to control the environmental conditions and prevent heat-related injuries.

==========
Example 4 | Record 220091180011 | Fatality | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
Employee was pinned between concrete structure and wall of hole.

Cause evidence: Employee was pinned between concrete structure and wall of hole.
Factor evidence:
Prevention rationale: Securing or repositioning the concrete structure and wall of the hole would prevent the employee from being caught in between them.

==========
Example 5 | Record 220063350035 | Days away | Early: 2000-2008
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: STRUCK_BY_MOVING_OBJECT
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Unsupported-Factor Safeguard Examples - 3 of 5

Narrative:
Employee was operating miner when a piece of rock fell hitting him on the neck. Rock measured 4'6" long, 2' wide and 2" thick. Doctor placed him on medical leave effective 11/27/2006.

Cause evidence: A piece of rock fell and hit the employee on the neck.
Factor evidence:
Prevention rationale: To prevent similar incidents, the rock should be properly secured or removed to avoid falling and injuring workers.

==========
Example 6 | Record 220140920009 | Fatality | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
A miner become buried from material sliding down the face of a bank.

Cause evidence: A miner became buried from material sliding down the face of a bank.
Factor evidence:
Prevention rationale: To prevent future incidents, the material should be stabilized or the movement of the material should be controlled.

==========
Example 7 | Record 220100710025 | Days away | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
EE was in process of changing out troughing idler on 30" conv. belt. While performing this task with a co-worker the idler tilted down and came in contact with the conv. frame. EE's right middle finger was mashed between the idler and the conv. frame. EE's right middle finger,last joint recieved approx. 11 stitches.

Cause evidence: EE's right middle finger was mashed between the idler and the conveyor frame while changing out the troughing idler.
Factor evidence:
Prevention rationale: Repair or replace the idler to ensure it remains stable during maintenance tasks and inspect all equipment before performing maintenance to prevent similar incidents.

Unsupported-Factor Safeguard Examples - 4 of 5

```
================================================================================
Example 8 | Record 220203530010 | Permanent disability | Recent: 2018-2026
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: STRUCK_BY_MOVING_OBJECT
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least
one support level is low; Unsupported factor conservatively replaced

Narrative:
Employee bent over to pick up a pogo stick laying on the ground near the installed vent curtain. A piece
of draw rock dislodged from the mine roof with height of 8'2" hitting employee's right hand resulting in
amputation to the right thumb.

Cause evidence: A piece of draw rock dislodged from the mine roof with a height of 8'2" hitting the
employee's right hand resulting in amputation to the right thumb.
Factor evidence:
Prevention rationale: Improving ground control or scaling to prevent rock dislodgement and ensuring
proper roof maintenance can reduce the risk of similar incidents.

================================================================================
Example 9 | Record 220122910012 | No lost time or other | Middle: 2009-2017
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: SLIP_TRIP_OR_FALL
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least
one support level is low; Unsupported factor conservatively replaced

Narrative:
EE was working in U/G Stores location changing security cameras. He was walking and slipped on an uneven
surface and strained his left knee. He went home and self treated the knee, it did not get better and he
was taken to the Dr. 10/11/2012 where he was given 2 cortisone injections and a prescription Pain
Medicine.

Cause evidence: EE slipped on an uneven surface while walking in U/G Stores location.
Factor evidence:
Prevention rationale: Improving the work surface condition and ensuring a level and safe walking surface
can prevent similar incidents.

================================================================================
Example 10 | Record 220200800005 | Fatality | Recent: 2018-2026
Analysis status: VALID_FOR_CAUSE_ANALYSIS_ONLY
Cause: STRUCK_BY_MOVING_OBJECT
Factor: NO_SEPARATE_FACTOR_STATED
```

Unsupported-Factor Safeguard Examples - 5 of 5

Prevention: INVESTIGATION_REQUIRED
Critical review:
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
Operator was priming Dredge pump with Jet line. When Operator closed the valve on Jet line, 2" Cam Lock broke and struck operator.

Cause evidence: When Operator closed the valve on Jet line, 2" Cam Lock broke and struck operator.
Factor evidence:
Prevention rationale: To prevent similar incidents, the stored energy in the system should be properly isolated or controlled before performing maintenance or operation.

Excluded and Critical-Review Examples - 1 of 5

These records contain an unresolved processing, routing, schema, cause-evidence, or alignment condition and are excluded from primary statistical conclusions.

==
Example 1 | Record 220041940006 | Days away | Early: 2000-2008
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: INSUFFICIENT_INFORMATION
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INSUFFICIENT_INFORMATION
Critical review: Cause is insufficient information; Cause evidence is not supported by narrative
Advisory review: No separate contributing factor stated; Allowed alternative prevention used instead of preferred mapping; At least one confidence value is low; At least one support level is low

Narrative:
While employee was pulling edge strips off of a stainless steel blank he felt a burning sensation in his right eye. His eye was rinsed out with saline solution and he was given antibiotics as a preventative measure.

Cause evidence: The narrative describes an employee experiencing a burning sensation in the right eye while pulling edge strips off a stainless steel blank. No specific mechanism of injury is described.
Factor evidence:
Prevention rationale: The narrative does not provide enough information to determine the cause or contributing factors, making it impossible to recommend specific prevention measures.

==
Example 2 | Record 220000380040 | Permanent disability | Early: 2000-2008
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: CAUGHT_IN_OR_BETWEEN
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Prevention: REPAIR_REPLACE_OR_INSPECT_EQUIPMENT
Critical review: Factor-prevention combination is outside allowed map
Advisory review:

Narrative:
EE WAS WORKING IN AN UNAUTHORIZED AREA. HE PUT HIS ARM BETWEEN TWO ROLLERS, TO CLEAN ONE OF THEM OFF. THE ROLLER CAUGHT THE SLEEVE ON HIS JACKET TWISTING AND PULLING OFF HIS ARM AT THE ELBOW.

Cause evidence: EE PUT HIS ARM BETWEEN TWO ROLLERS TO CLEAN ONE OF THEM OFF. THE ROLLER CAUGHT THE SLEEVE ON HIS JACKET TWISTING AND PULLING OFF HIS ARM AT THE ELBOW.
Factor evidence: EE PUT HIS ARM BETWEEN TWO ROLLERS, TO CLEAN ONE OF THEM OFF. THE ROLLER CAUGHT THE SLEEVE ON HIS JACKET TWISTING AND PULLING OFF HIS ARM AT THE ELBOW.
Prevention rationale: Repair or replace the rollers to ensure they are in good working condition and do not catch clothing or body parts.

==
Example 3 | Record 220142730019 | Days away | Middle: 2009-2017
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: STRUCK_BY_MOVING_OBJECT

Excluded and Critical-Review Examples - 2 of 5

Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Prevention: ISOLATE_OR_CONTROL_STORED_ENERGY
Critical review: Factor-prevention combination is outside allowed map
Advisory review:

Narrative:
Tube was latched onto and pulled out of the hole. Safety pin was applied to the overshot before hoisting core tube out of hole. As driller and helper began lowering tube one of the two lifting dog teeth broke off completely allowing tube to fall to the deck. The core tube landed on an block on the drill deck, which caused the tube to kick up and strike helper in left eye brow.

Cause evidence: As driller and helper began lowering tube one of the two lifting dog teeth broke off completely allowing tube to fall to the deck.
Factor evidence: The core tube landed on an block on the drill deck, which caused the tube to kick up and strike helper in left eye brow.
Prevention rationale: To prevent the lifting dog teeth from breaking and the tube from falling, the stored energy in the lifting mechanism should be isolated or controlled. This could involve securing the lifting mechanism or ensuring the lifting teeth are in good condition.

==========
Example 4 | Record 220063130057 | Fatality | Early: 2000-2008
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: PRESSURE_OR_STORED_ENERGY_RELEASE
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review: Cause evidence is not supported by narrative
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Accident happened when a cut of coal was blasted in breast. From preliminary report: According to the witness, he heard three explosions. Two were normal. The concussion from the third explosion was excessive. The victim was eventually found at the bottom of #19 chute.

Cause evidence: The accident occurred due to an excessive concussion from a third explosion, indicating the release of stored energy.
Factor evidence:
Prevention rationale: To prevent future incidents, the stored energy from the blasting process should be better controlled or isolated.

==========
Example 5 | Record 220051220051 | Fatality | Early: 2000-2008
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: INSUFFICIENT_INFORMATION
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review: Model response failed validation; Cause is insufficient information; Cause evidence is not supported by narrative; Factor-prevention combination is outside allowed map

Excluded and Critical-Review Examples - 3 of 5

Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Technician was attempting to repair a steering problem on a wheel loader. He replaced steering pilot valve, started engine, raised bucket and attempted to steer machine. When it would not steer to the right, he dismounted machine without shutting off engine. While attempting to remove steering shaft, the steering activated and steered to the right crushing the victim.

Cause evidence:
Factor evidence:
Prevention rationale: The model response could not be validated.

==========
Example 6 | Record 220002210014 | Permanent disability | Early: 2000-2008
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Prevention: REPAIR_REPLACE_OR_INSPECT_EQUIPMENT
Critical review: Factor-prevention combination is outside allowed map
Advisory review:

Narrative:
ON THE LOCATION WHERE SOME OLD DRYERS HAD BEEN TAKEN OUT FOR NEW DRYERS; NEW CHUTES WERE BEING INSTALLED. ONE OF THE CHUTES TURNED OVER AND THE EMPLOYEE TRIED TO HOLD IT FROM HITTING THE FLOO R HARD ENOUGH TO DAMAGE THE CERAMIC LINING. HE DID NOT EXPERIENE PAIN AT THIS TIME, BUT HAD A KNOT ON HIS GROIN AND PAIN THE NEXT DAY.

Cause evidence: ONE OF THE CHUTES TURNED OVER AND THE EMPLOYEE TRIED TO HOLD IT FROM HITTING THE FLOOR HARD ENOUGH TO DAMAGE THE CERAMIC LINING.
Factor evidence: NEW CHUTES WERE BEING INSTALLED ON THE LOCATION WHERE SOME OLD DRYERS HAD BEEN TAKEN OUT FOR NEW DRYERS.
Prevention rationale: Repair or replace the chutes to ensure they are stable and properly installed to prevent them from turning over.

==========
Example 7 | Record 220193400024 | Permanent disability | Recent: 2018-2026
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: MANUAL_HANDLING_OR_OVEREXERTION
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review: Cause evidence is not supported by narrative
Advisory review: Narrative may be truncated; No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
EE was stacking 50# sand bags on a pallet. When EE turned to place a bag on the pallet had a sharp pain in the abdomen. Later in the day EE had a lump protruding from the same area. EE can push the lump back

Excluded and Critical-Review Examples - 4 of 5

in. *On 03/11/2020 EE saw a doctor & was diagnosed with a hernia. EE plans to have the hernia surgically repaired. The doctor issued a Return to Work slip with No Limitations.

Cause evidence: EE was stacking 50# sand bags on a pallet and experienced a sharp pain in the abdomen after turning to place a bag on the pallet.
Factor evidence:
Prevention rationale: The task of stacking heavy sand bags can lead to overexertion and the development of a hernia. Redesigning the workspace to reduce the need for heavy lifting or providing mechanical assistance could prevent such incidents.

==========
Example 8 | Record 220242530009 | Days away | Recent: 2018-2026
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: CAUGHT_IN_OR_BETWEEN
Factor: UNSTABLE_OR_IMPROPERLY_POSITIONED_MATERIAL
Prevention: REPAIR_REPLACE_OR_INSPECT_EQUIPMENT
Critical review: Factor-prevention combination is outside allowed map
Advisory review:

Narrative:
Employee removed a pin from the bottom of a hatch door to raise up the door. As employee was raising the door, the door came down and struck right index finger pinching the finger between the door and frame resulting in a laceration requiring sutures to close. Employee returned to work same day on full duty.

Cause evidence: As employee was raising the door, the door came down and struck right index finger pinching the finger between the door and frame.
Factor evidence: Employee removed a pin from the bottom of a hatch door to raise up the door.
Prevention rationale: Repair or replace the hatch door to ensure it functions properly and does not unexpectedly come down and strike the employee.

==========
Example 9 | Record 220122060028 | Fatality | Middle: 2009-2017
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: LOSS_OF_CONTROL_OF_TOOL_OR_EQUIPMENT
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review: Cause evidence is not supported by narrative
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low

Narrative:
Witness saw the water truck traveling downhill on Ridge road & then saw a cloud of dust. Truck traveled downhill(9% grade)struck berm on the right side of road ejected operator changed direction & traveled across road striking the berm/slope on the left side of the road. The truck overturned & came to rest. The victim did not appear to use his seat belt. Minor damage to truck.

Cause evidence: The water truck traveled downhill on a 9% grade and struck a berm, causing it to change direction and strike another berm, resulting in an overturn.

Excluded and Critical-Review Examples - 5 of 5

Factor evidence:
Prevention rationale: The truck should be inspected and repaired to ensure it is in proper working condition to prevent loss of control on a steep grade.

==
Example 10 | Record 220160430005 | Permanent disability | Middle: 2009-2017
Analysis status: EXCLUDED_OR_REVIEW_REQUIRED
Cause: CAUGHT_IN_OR_BETWEEN
Factor: NO_SEPARATE_FACTOR_STATED
Prevention: INVESTIGATION_REQUIRED
Critical review: Cause evidence is not supported by narrative
Advisory review: No separate contributing factor stated; At least one confidence value is low; At least one support level is low; Unsupported factor conservatively replaced

Narrative:
Employee was in the process of securing a barge to be loaded, using a chain and clevis. The wind shifted and the barge moved up-river. EE had EE's hand on the chain when this occurred. The accident resulted in an amputation of EE's left index finger (amputated at local hospital).

Cause evidence: The wind shifted and the barge moved up-river, causing the employee's hand to be caught in the chain.
Factor evidence:
Prevention rationale: Improving the securing method to prevent the barge from moving unexpectedly would reduce the risk of the employee's hand being caught in the chain.

Interpretation and Limitations

1. The sample is balanced by severity. It does not represent the natural severity distribution of the complete MSHA dataset.

2. Time trends describe narrative-category shares, not exposure-adjusted accident rates.

3. The primary cause represents the reported event mechanism, not a verified root cause.

4. Prevention themes are preliminary analytical categories, not approved corrective actions.

5. Short narratives may describe what occurred without explaining why.

6. NO_SEPARATE_FACTOR_STATED is a valid result when a factor is not supported.

7. Block 3 will run the frozen system on 100 additional unseen narratives and create the independent human-review workbook.

8. Block 4 will compare the completed human labels with the model labels and generate the final validation report.

Technical Appendix

```
Model: Qwen/Qwen2.5-7B-Instruct
Prompt version: MSHA_BLOCK1C_FROZEN_FOR_BLOCK2_V3.0
Unseen records processed: 2,000
Block 1 IDs excluded: 1,000
Target per severity group: 400
Random state: 84
Maximum retries: 4
Maximum generated tokens: 750
Minimum narrative words: 8
Truncation threshold: 383

PDF: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data Science/MSDS 453-
AI and Natural Language/Final Project/MSHA_Block2_Unseen_Narrative_Analysis_Report.pdf
Results CSV: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block2_Unseen_Results.csv
Checkpoint: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block2_Unseen_Checkpoint.jsonl
Block 2 IDs: /mnt/batch/tasks/shared/LS_root/mounts/clusters/dvandyk2/code/Users/dvandyk/NW Data
Science/MSDS 453- AI and Natural Language/Final Project/MSHA_Block2_Unseen_Record_IDs.json
```